1 Project Overview

SANAD Training System / Training Application Module

OBJECTIVE

Deliver a production-ready, standalone RESTful API for handling course applications.

With high standards of

Security | Quality | Maintainability

CORE DELIVERABLES

1. Clean, modular architecture

2. End-to-end integration of

- Authentication
- Input Validation
- Audit Logging
- Persistence
- Automated Testing

2 Technical Stack

Tools and libraries behind the module

RUNTIME & FRAMEWORK

Node.js & Express.js

Fast, minimal HTTP layer for the REST API.

DATABASE & ORM/ODM

MongoDB + Mongoose

Schema modeling for the application data.

AUTHENTICATION

JWT

JSON Web Tokens for stateless identity verification.

VALIDATION

Zod / Joi

Schema validation for request payload sanitization.

TESTING

Jest & Supertest

Robust integration testing of every endpoint.

3 Architecture & Security Design

From request to persisted record

REQUEST LIFECYCLE

1 Authentication
Middleware

Validates `Bearer JWT` and populates identity context into `req.user`.

2 Validation
Middleware

Evaluates the body against strict schemas; returns `400 Bad Request` on failure before reaching controllers.

3 Controller
Logic

Executes business operations and manages database persistence.

AUDIT LOGGING

Auto-injects createdBy and updatedBy fields linked to the authenticated user ID.

Enables Mongoose timestamps (createdAt, updatedAt) for operational traceability.

4 API Endpoints & Postman Integration

What the module exposes and how to try it

EXPOSED ENDPOINTS

POST `/api/applications`

Submit a new training application.

GET `/api/applications`

Fetch applications owned by the authenticated user.

POSTMAN SETUP

Pre-configured collection

`sanad-api.postman_collection.json`

Environment variables

`{{baseUrl}}` `{{token}}`

Includes pre-built test scripts for **success paths** and **error handling**.

5 Testing & Quality Assurance

Jest + Supertest integration coverage

TEST COVERAGE MATRIX

PASS

201 / 200

Created / OK

Validates successful creation and fetching.

PASS

401

Unauthorized

Verifies access denial when the JWT token is missing or invalid.

PASS

400

Bad Request

Validates proper rejection of invalid payloads.

OUTCOME

A **clean, self-contained, and fully tested** backend module, ready for seamless integration into the SANAD ecosystem.